LaPierre Updated Summarized GRIN Project Proposal

Linking Seasonal Timing of Fire, Soil Disturbance, and Seed Arrival to Post-Fire Recovery and Restoration

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How is community assembly influenced by the seasonal timing of fire, soil disturbance, and or dispersal limitations?

Seasonal Timing Influence on Soil Disturbance

Soil disturbance, like fire, is a driving evolutionary force of biological communities. Weedy plant species, referred to as ruderals, have life history strategies and traits that allow them to respond positively to soil disturbance[1]. Ruderals can rapidly invade and become abundant in areas after soil disturbance[2]. Prior to humans, intense soil disturbances would have likely been small, infrequent, and seasonal, resulting from the wallowing of megafauna as they migrated[3,4]. Today, anthropogenic soil disturbances, such as agriculture and silviculture activities, are ubiquitous, and occur at greater intensities, frequencies, and scales, as well as during all seasons. Anthropogenic soil disturbances have been shown to increase the abundance and range of many ruderal species[5], making their ecological roles unclear[6].

The influx of certain ruderal species across disturbed sites are known to inhibit restoration by increasing competition, limiting seed dispersal, and altering fire behavior[7–10]. Perennial forb ruderal species, such as *Eupatorium compositifolium* or "dogfennel"[11] for example are also thought to affect restoration negatively by burning less intensely than other obligate species of healthy fire-adapted habitats[9]. Field observations suggest however that certain ruderal grasses, such as *Andropogon virginicus*, encourage the movement of fire across restoration sites, which may overtime positively influence the recruitment of obligate savanna species. It is not clear however what factors lead to higher densities of differing types of ruderal species, e.g. forbs or grasses, across disturbed sites.

In general, soil disturbance is thought to create abiotic conditions favorable for the recruitment of ruderal species during community assembly, including increased available recruitment space[10], influx of nutrient and microbial activity[12], along with light and moisture conditions favorable for germination[10]. The seasonal timing of soil disturbance is in turn known to alter the recruitment patterns of different ruderal species across old-field environments primarily because species germinate under different seasonal timing patterns[13–15]. Large-seeded ruderal species for example have been shown to be more adversely affected by tilling disturbance during favorable germination conditions than smaller seeded species[10]. Consequently, the recruitment of certain species early after soil disturbance activities have been shown to have priority effects on community assembly pathways[16], which can alter timelines on management goals and objectives.

Understanding the influence that the seasonal timing of soil disturbance has on early recruitment will provide restoration ecologists with a more complete understanding of how community assembly can be potentially manipulated. Tracking the recruitment and the changes to plant communities' overtime after soil disturbance will also provide useful information concerning patterns in ruderal development and potential influence over one another. Additionally, by testing the effect that the seasonal timing of soil disturbance has on ruderal recruitment of sites we can potentially provide information on how to prevent or manipulate the influx of certain types of ruderal species over others. Information concerning the influx of unfavorable ruderal species would be very useful for southeastern land mangers endeavoring to restore biodiverse fire-prone savannas.

Primary Question:
How does seasonal timing of soil disturbance influence recruitment of ruderal species and plant community assembly?

Treatments:

Winter Till Disturbance (T_W)

Spring Till Disturbance (T_{Sp})

Control No Till Disturbance (C)

2 Treatments + Control; 4 Replications Per 2 Blocks; N = 8; Total Plots = 24

Hypotheses:

T_W – Winter till disturbance will result in the greatest first-year recruitment of woody-ruderal species.

- Primarily due to the timing proximity to recent fall mass seeding event, and favorable abiotic recruitment conditions. Second year abundance of woody-ruderals will increase and be the highest among treatments.

T_{Sp} – Spring till disturbance will result in lower first-year recruitment of any species and high presence of bare-ground.

- Primarily due to harsh abiotic recruitment conditions (max temperature).

C – No till disturbance control will result in very similar plant community composition and abundances across years.

The Seasonal Timing Interactions Between Fire & Soil Disturbance

Regarded as a filter in community assembly, fire is known to alter plant community composition overtime by increasing the abundance of species better adapted for fire[17–19]. Land managers frequently manipulate the seasonal timing of fire in order to maintain grass- and forb-dominated communities, while limiting the growth of woody shrubs and ruderal species[20] for the benefit of preserving diversity in savanna ecosystems[19,21]. Many researchers suggest that ecosystems benefit more broadly when the seasonal timing of fire mimics the historical fire season[22–27]. In the southeast U.S.A., prior to human disturbances, lightning-ignited fires were most common during the transition between drier and wetter seasons (April-June)[18,28–30]. Today a large proportion of prescribed fires, due to practical constraints, take place outside of this historical fire season, potentially altering the efficacy of some land management and restoration efforts[19,27,30,31]. The effects and value of fire seasonality in maintaining and restoring plant communities, however, remains complicated due to urban-interface conflicts, management resource limitations, climate change, and insufficient restoration research[30,32].

Research has shown thus far that the seasonal timing of fire is important as it plays a prominent role in stimulating flowering, seeding, and mass synchronicity of the two, across southeastern ecosystems[33–37]. Mass flowering and seeding among southeastern savanna species is known to occur in fall months (September-November) – primarily following late-spring (May-June) to early summer (June-July) fires[23,35,38,39]. In fire-prone systems of north Florida, the germination of most fall seeding species has been documented to take place predominantly during winter months (December-February), when precipitation and lower maximum temperatures conducive for

seedling germination are known to occur[40–42]. Observations suggests that most ruderal species, many of which also seed during fall months, vastly outproduce seeds compared to most obligate savanna species when fire is absent. It has also been documented though that ruderal seeds often exist in high quantities within the seedbank of undisturbed fire frequented savannas, despite low aboveground densities and little recruitment[40]. Furthermore, while it is known that soil disturbance fosters ruderal recruitment[2,43], it is not clear exactly how fire negatively influences it.

The seasonal timing of fire early in restoration may potentially play a substantial role in altering plant community composition and the trajectory of restoration outcomes. Research has demonstrated that winter-dormant season fires are less effective at reducing the stem density of woody shrubs than spring- or summer-growing season fires[44]. Consequently, early development of weedy fire impeding ruderals after soil disturbance may also in turn be less affected by dormant-winter fires than spring-growing fires. Preliminary research has shown that perennial-forb ruderal species such as dogfennel are negatively influenced by spring-growing season fires than winter-dormant season fires. Consequently, the timing of soil disturbance and fire filtering events may drive different non-stochastic patterns of plant community composition and structure overtime. Furthermore, the interactions between both the seasonal timing of fire and soil disturbance may also interact to create positive or negative feedback cycles that influence restoration outcomes overtime.

Overall, little information exists concerning how fire influences ruderal recruitment across recently disturbed sites. Detecting the influence of early fire effects could provide restoration ecologists with a useful framework to understand community assembly pathways after soil disturbance. Moreover, fire and soil disturbances are commonly used across the southeastern U.S.A. by land managers for a variety of purposes. Understanding how fire influences the abundances of different types of ruderal species early on after soil disturbance may help inform land managers and stakeholders in their decision-making process across sites. Information such as this would also be valuable for restoration activities across the southeast, where restoration of fire prone ecosystems is becoming increasingly common, and the influx of fire impeding ruderals are a major issue hampering timelines[8,45].

<u>Primary Questions:</u>

Does fire prior to occurrence of soil disturbance influence later community assembly?

<u>Treatments:</u>

*No-burn and No-Till control treatments will be utilized from soil disturbance project.

Winter Tiller Disturbance – Winter-Fire (T_W-F_W)

Spring Tiller Disturbance – Winter-Fire (T_{Sp}-F_W)

No Tiller Disturbance – Winter-Fire (T_C-F_W)

Winter Tiller Disturbance – Spring-Fire (T_{Sp}-F_{Sp})

Spring Tiller Disturbance – Spring-Fire (T_{Sp}-F_{Sp})

No Tiller Disturbance – Spring-Fire (T_C-F_{Sp})

Winter Till Disturbance (T_W)

Spring Till Disturbance (T_{Sp})

Control No Till Disturbance or Fire (C)

9 Treatments Total; 4 Replications Per 2 Blocks; N = 8; Total Plots = 96

Hypotheses:

T_W-F_W: Winter tiller disturbance and winter fire disturbance will result in the greatest coverage of forb perianal ruderals such as dogfennel.

- Primarily due to timing proximity to recent mass seeding and production of favorable recruitment abiotic conditions.
- Second year abundance and dominance of forb perennial ruderal species will increase and be high among treatments.
- Winter fire will have little effect on forb perennial ruderal species due to fire timing in relation to carbohydrate reserves thus third-year abundances will continue to increase compared to T_W treatment.

T_{Sp}-F_W: Spring tiller and winter fire disturbance will result in low to moderate first-year recruitment of perennial forb ruderals and moderate recruitment of grass ruderal species.

- Primarily due to prolonged timing proximity to recent mass seeding and less favorable ruderal recruitment conditions.
- Second year abundance and dominance of grass ruderal species will increase and be high among treatments.
- Winter fire will have little effect on perennial forb ruderal species due to fire timing in relation to carbohydrate reserves thus third-year abundances will continue to increase compared to T_W treatment.

T_C-F_W: Winter fire and no tilling disturbance will result in very similar plant community composition and abundances across years.

- Winter fire will slightly enhance total species recruitment and species richness two-year post-tilling over T_C treatment alone.

T_{Sp}-F_{Sp}: Spring tilling coupled with spring fire will enhance the recruitment of grass ruderal species.

- Primarily due to prolonged timing proximity to recent mass seeding, and favorable ruderal recruitment conditions.
- Second year abundance and dominance of grass ruderal species will increase.
- Spring fire will reduce abundance of perennial forb ruderal species.

T_C-F_{Sp} : No tilling disturbance and spring fire will result in very similar plant community composition and abundances across years.

- Spring fire will slightly enhance total species recruitment and species richness three-year post-tilling over T_{Con} and T_W treatments.

Seasonal Timing of Fire Effects on Seeding Restoration

Historically, the reassembly of many plant communities following soil and fire disturbances likely occurred and evolved in conjunction with adequate seed availability (e.g. viable seedbank or adjacent population sources). Restoration sites however are often seed-limited, due to frequent and intense soil disturbances[46,47]. Direct seeding following large-scale disturbances is becoming more

common in restoration[48–50], where it is viewed as a way of "jumpstarting" recovery efforts[46]. The use of direct seeding to restore sites is continuing to increase as human disturbances expand[51,52], seed costs decrease, and new techniques for seeding are developed (e.g. strip seeding, aerial seeding)[53]. In short, direct seeding is an increasingly common tool used in restoration of savannas to increase seed input where the native obligate species pool is limited.

The use of direct seeding in restoration of southeastern plant communities is often coupled with the use of fire which helps maintain grass and forb dominated plant communities[29,54]. The seasonal timing of fire though, is known to influence the growth and reproduction of many obligate savanna species[19,30,32]. The seasonal timing effect of fire on recruiting species during early restoration however is not as well understood. Preliminary research suggests weedy ruderal species that establish after soil disturbance events are less affected by winter fires than transition season fires[19,44]. Increased establishment of certain perianal forb ruderal species is thought to negatively influence the restoration of obligate savanna groundcover species via competitive exclusion, priority effects, altering fire filtering, and limiting seed dispersal[10,55]. The seasonal timing of fire may also directly affect restoration pathways by inducing flowering of newly established groundcover species that go on to then increase further potential of new additional recruitment. Additionally, there might also be unknown competitive or nursing interactions between different recruited species that may also affect community development overtime. Overall, a lack of information exists concerning the seasonal timing effects of fire across seeded restoration sites of savanna systems.

<u>Treatments</u>

*Seeding treatments are essentially a replicate of tilling treatments with the addition of seeding.
*No-burn and no-till control treatments will be utilized from soil disturbance project.

Seed Addition – No Fire (S)

Seed Addition – Winter Fire ($S\text{-}F_W$)

Seed Addition – Spring Fire ($S\text{-}F_{Sp}$)

2 Treatments + Control Total; 2 Replications Per 3 Fire Blocks; N = 8; Total Plots = 24

<u>Hypotheses:</u>

S: Seed addition of obligate sandhill savanna species recruit with moderate ruderal recruitment.

- Third year ruderal abundances increase, seeded species are outcompeted and displaced.

$S\text{-}F_W$: Seed addition of obligate sandhill species recruit with moderate ruderal recruitment.

- Third year ruderal abundances increase, seeded species are slightly outcompeted and displaced, dormant-winter season fire does little to reduce ruderal domination.

$S\text{-}F_{Sp}$: Seed addition of obligate sandhill species recruit with moderate ruderal recruitment.

- Third year ruderal abundances decrease, seeded species outcompete ruderal species, with growing-spring season fire effectively reducing ruderal abundances.

Sandhill Seeding Species Selection

Family	Species	Plant Type
Poaceae	*Andropogon ternarius*	Grass
Poaceae	*Anthaenantia villosa*	Grass
Poaceae	*Aristida beyrichiana*	Grass
Asteraceae	*Carphephorus corymbosus*	Forb
Poaceae	*Ctenium floridanum*	Grass
Asteraceae	*Liatris tenuifolia*	Forb
Asteraceae	*Pityopsis graminifolia*	Forb
Asteraceae	*Solidago odora*	Forb

Methods

Experimental Design

Among the three projects detailed above a total of 98 plots ($1m^2$) will be established across two replicated (30m x 100m) experimental blocks consisting of three ($18m^2$) fire treatment areas (No-Fire, Winter-Fire, Spring-Fire). Fire treatment areas will feature a 10m buffer between one another; both experimental blocks will be placed no closer than 10m from one another. All plots will be established with angle-iron corners featuring center points located on the southern boundary of each plot. All plots will feature both a 1m buffer surrounding each plot along with an additional 2m placement buffer from other plots.

In order to eliminate potential confounding matrix effects, the experiment will take place in the middle of a field dominated by bahiagrass (*Paspalum notatum*), featuring low species richness, and zero percent canopy coverage. In order to successfully remove the dominant exotic bahiagrass, herbicide using 4% glyphosate, at a rate of five quarts per acre, will be applied to all treatment areas four months preceding all treatments. In December a prescribed burn will take place across both experimental blocks which will clear out the treatment areas of dead bahiagrass. Plots will then be placed into tilling or seeding treatment sub-blocks within fire treatment blocks.

Tilling treatments will be performed with a rear-tine rototiller and at a depth of 10cm. The use of a rototiller in this project is preferred primarily because it is portable, it minimizes soil disturbances elsewhere on the site, and it is easily controllable. The 10cm tilling depth was selected because it most closely matches common anecdotally referenced widespread silvicultural soil disturbance depths in this region, along with the successful recommendations across seeding studies. The biomass of all residual species will be removed directly after tilling to eliminate the effects of residual species that include bahiagrass. All soil disturbance treatments, and seed treatments, will be applied one to two days prior to a significant predicted rainfall event of one inch or more.

Seed treatments will be conducted by adding twelve locally sourced sandhill selected species. Seed viability of each species will be tested prior using a tetrazolium test. Viability rates will then be used to adjust base seeding rates, which will be standardized at 25 seeds per $1m^2$ plot based on selected studies[48,56–59]. Seeds will be sown after tilling using a hand-rake at a 1cm depth. A 50kg sod roller will be used to cover seeds immediately after sowing to prevent seed predator losses and break seed dormancy as per established seeding methods[48,54,60]. All seeding treatments will take place in in the month of December 2020 with 48 hours of significant rainfall. December sowing date was chosen because in the southeast, many obligate savanna species are known to set

seed in the late fall directly following spring or summer fires, and then germinate during later winter months following rain brought in by cold fronts [40,48,61].

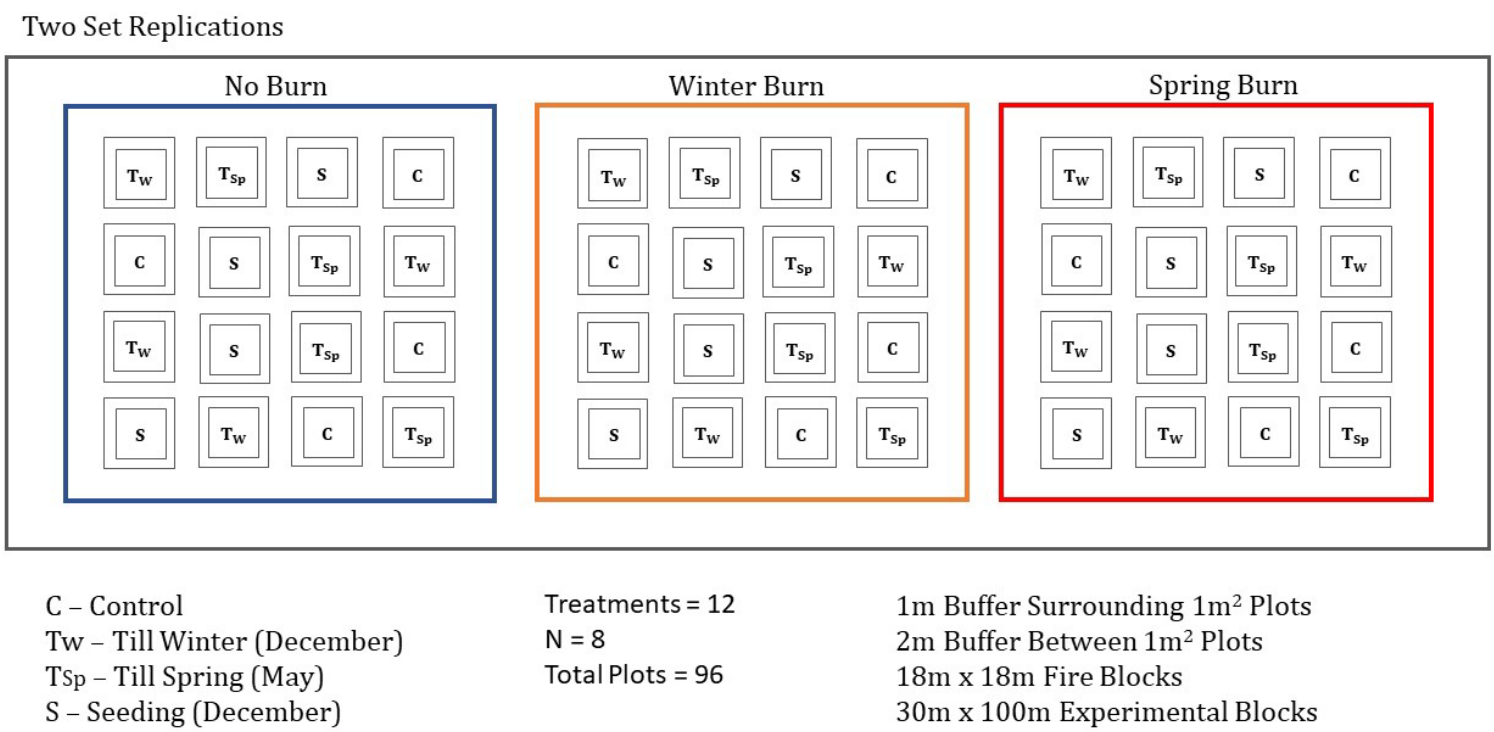
Finally, all fire treatments will take place in 2022 on a two-year fire cycle that will continue until the study is no longer deemed necessary. Ignition of dormant-winter fire treatment areas will take place in the month of December, while ignition of growing-spring fire treatment areas will take place during the month of May. Burns in fire treatment areas will take place with mowed fire lines surrounding them as opposed to tilled fire lines. Burn conditions and ignition techniques will be provided and monitored across sites.

Measurements

Plant community composition will be assessed using coverage and abundance data conducted prior to treatments (October 2020) as well as each year beginning in the fall (October-November) when most species on site are in flower. Percent cover of each species will be estimated at a $1m^2$ level in subplots using 10 cover values (1 = trace, 2 = <1%, 3 = 1-2%, 4 = 2-5%, 5 = 5-10%, 6 = 10-25%, 7 = 25-50%, 8 = 50-75%, 9 = 75-95%, 10 = >95%). Abundance data will be performed across $1m^2$ plots using an internal cardinal grid system (NE, NW, SE, SW) to accurately count all individual plants, stems, and new seedlings. Monthly aerial plot photos will be taken using a high-resolution camera on a tripod with an arm extender, at 1m height in the center of each plot. Photos will be utilized with software to assess and develop coverage and micro-geospatial relationships and patterns of different treatments overtime.

The seedbank will also be measured by sampling the soil of the buffer area of each plot, at a depth of 5 cm and 10 cm prior to treatments. Soil seedbank samples will then be germinated via lab bench using the following methods described in Sharma(2018)[62]. Ruderal seedlings will be counted and differences in seedbank samples across different treatments will be assessed and analyzed to account for any potential block effects.

Figure 1A: Seasonal soil-disturbance, fire, & seeding project experimental design.



Statistical Methods

In order to test these predictions, permutational multivariate analysis of variance (PERMANOVA) will be used with the adonis function in the package vegan. To visually compare differences in plant community composition non-metric multidimensional scaling (NMDS) will be utilized using the function metamds from the vegan package. Prior to use of NMDS, a calculation of Bray-Curtis dissimilarities will be conducted using the vegdist function from the package vegan (Myers et al. 2015). All multivariate analysis is done using R studio version 3.6.1. Packages used to perform analysis and graphs in R studio include: ggplot2, reshape2, gridExtra, tidyverse, vegan, grid, and labdsv.

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